AWS Fargate

AWS Fargate[35] is a new platform from Amazon that puts a serverless spin on their ECS and EKS services. Both ECS and EKS, by default, require some manual steps to create the EC2 instances that will run the containers. Fargate is a drop-in replacement compute engine for ECS and EKS that removes the need to think about and manage server instances.

How does this work in practice? You specify your containers, as well as config regarding how the containers should be managed at runtime. Based on the CPU and memory parameters you set, Fargate will automatically scale your containers up or down as necessary, based on load; you don't need to care where or how they're run.

Fargate containers can be used to run one-off, short-lived tasks—much like AWS Lambdas. In addition, Fargate is currently the only Serverless offering that supports traditional-style applications and microservices.

Although Fargate removes a lot of the orchestration burden of running containers in production, as with AWS EKS, you still have to have some interaction with underlying AWS services to configure your networking and specify security policies.

Pricing for Fargate is based on virtual CPU and memory usage, and is currently anything from two to six times more expensive than running an EC2 instance. The gap is even wider when you consider that an EC2 instance is capable of running multiple containers. So, although an interesting service to watch, the price needs to come down before it will gain real momentum.

Microsoft Azure Container Instances

Microsoft Azure Container Instances (ACI)[36] is a much more limited form of serverless for containers. It can't be used to run a standard containerized application like our Rails app we've been building throughout the book. Instead, it's intended for applications that have built from the ground up to be Serverless; it's suitable for discrete tasks (such as data processing) or event-driven applications.

As with Fargate, pricing is based on virtual CPU and memory usage, and the two services cost roughly the same.

35. https://aws.amazon.com/fargate/
36. https://azure.microsoft.com/en-gb/services/container-instances/